

Design and Development of a Height-Adjustable Washbasin for Wheelchair Users

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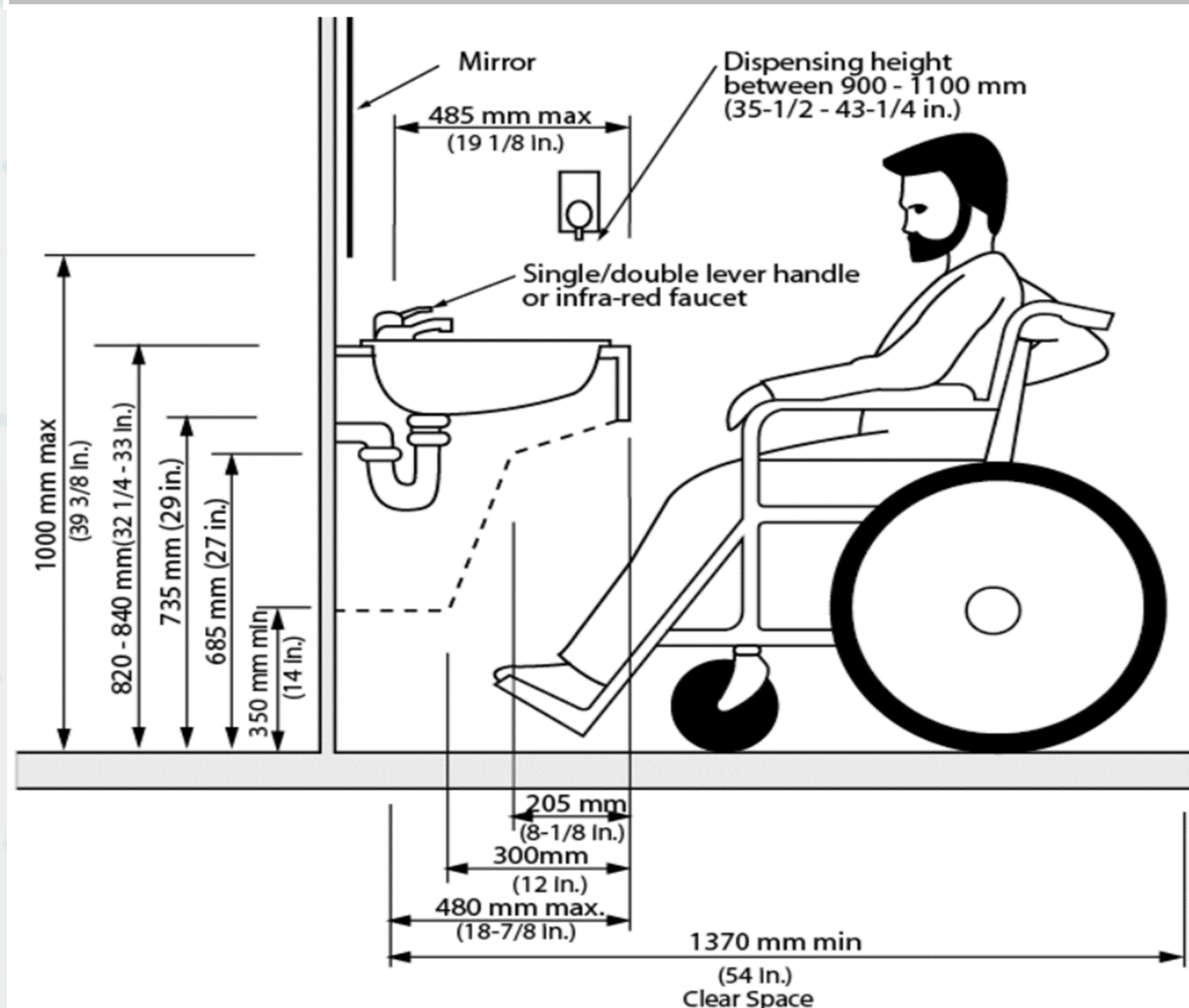


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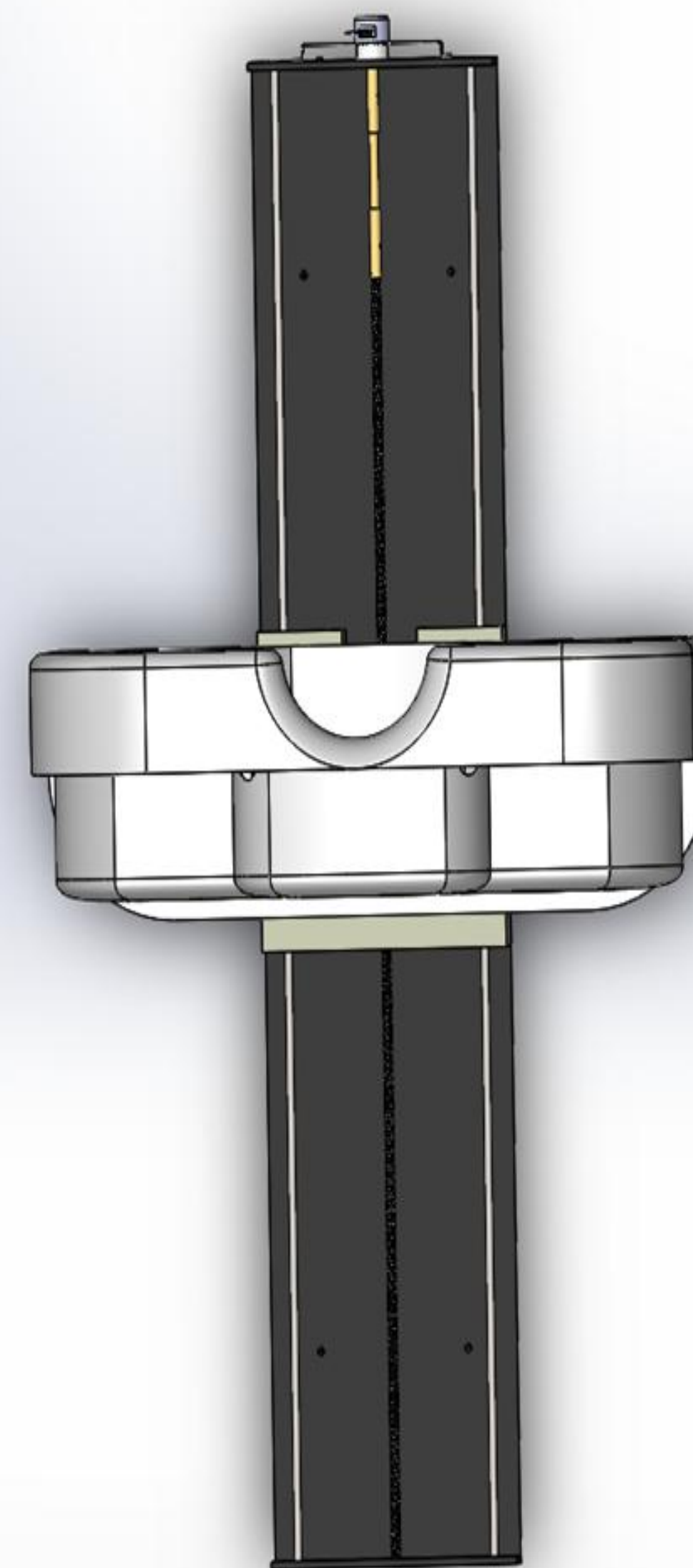
About the Project

The proposed system is a height-adjustable washbasin designed to improve accessibility for wheelchair users in public and residential bathrooms. The mechanism converts rotary motion generated by a DC motor into controlled linear motion using a lead screw assembly. This allows the washbasin height to be adjusted according to the user's needs, providing improved comfort, accessibility, and independence.



Project Aims:

- Improve accessibility for wheelchair users.
- Provide comfortable use for people with different mobility requirements.
- Develop a reliable height-adjustment mechanism.
- Offer a cost-effective and practical solution.
- Promote inclusive bathroom design.



Technologies Used:

- 1- Linear actuators (Lead screw)
- 2- Rotary actuators (DC motor)
- 3- Microcontroller (Arduino UNO)
- 4- Software programs (SolidWorks, Proteus)
- 5- Sensors (PIR sensor, limit switch)
- 6- Valves



Results:

- Successfully designed and implemented a functional prototype.
- Achieved smooth vertical movement using a lead screw mechanism.
- Improved accessibility and usability for wheelchair users.
- Demonstrated reliable operation during testing.
- Developed using affordable and readily available components.

References

1. Resnick, Robert and Halliday, David (1966), Physics, Section 3-4 .
2. Robotics Research Group. "Joint Types". University of Texas at Austin. Archived from the original on 2009-03-11. Retrieved 2009-02-04.